

Operating System (ICE 227)

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Outline

- Operating System Services

Operating System Services

- Operating systems provide an environment for execution of programs and services to programs and users
- **User interface** -Almost all operating systems have a user interface (UI).
- **Program execution** -The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally (indicating error)
- **–I/O operations** -A running program may require I/O, which may involve a file or an I/O device

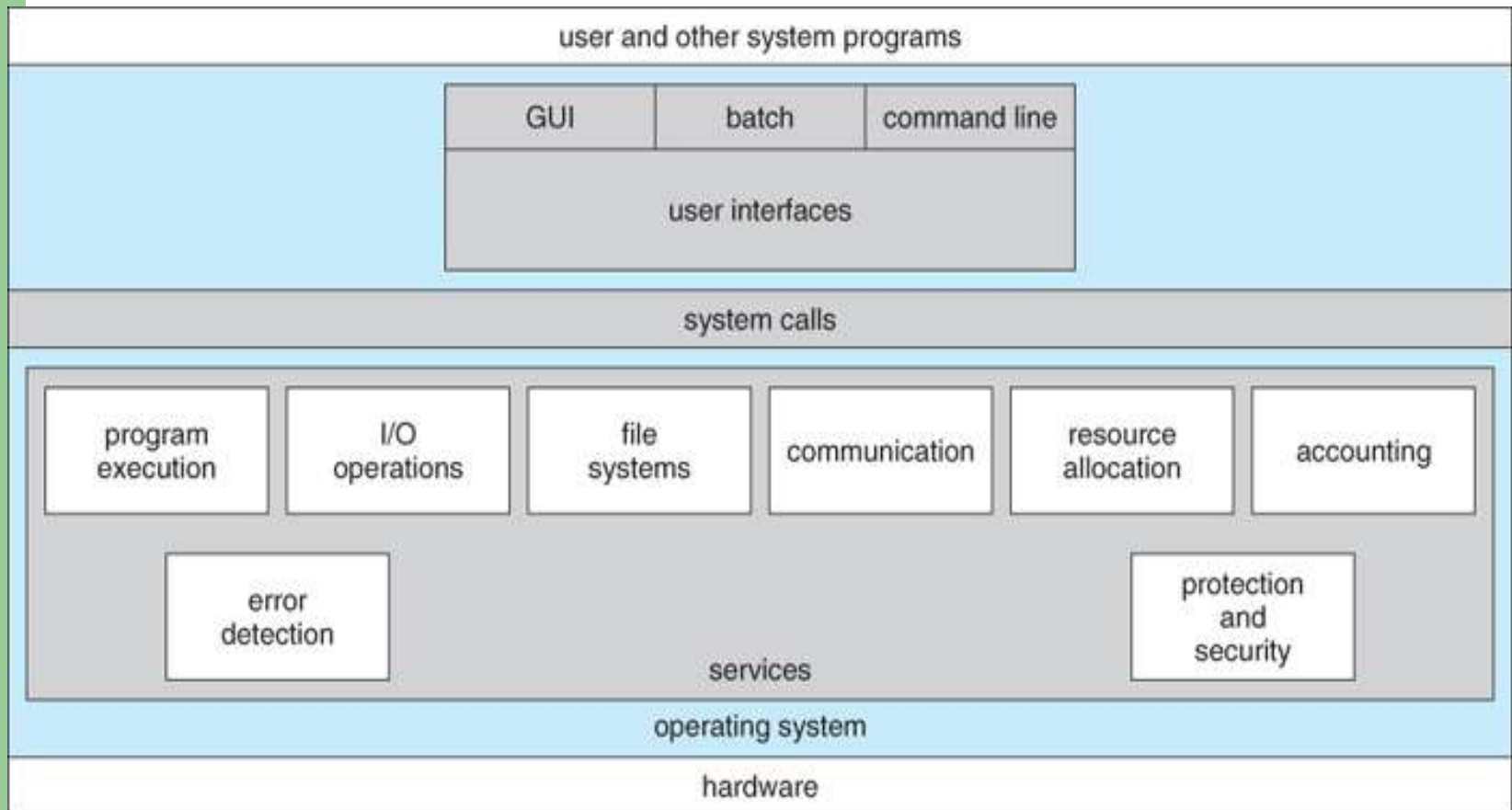
Operating System Services

- **File-system manipulation** -Programs need to read and write files and directories, create and delete them, search them, list file Information, permission management.
- **Communications**—Processes may exchange information, on the same computer or between computers over a network
- **Error detection** –OS needs to be constantly aware of possible errors

Operating System Services

- Another set of OS functions exists for ensuring the efficient operation of the system itself via resource sharing
- **Resource allocation** -When multiple users or multiple jobs running concurrently, resources must be allocated to each of them
- **Accounting** -To keep track of which users use how much and what kinds of computer resources
- **Protection and security** -
 - Protection** involves ensuring that all access to system resources is controlled
 - **Security** of the system from outsiders requires user authentication, extends to defending external I/O devices from invalid access attempts

Operating System Services



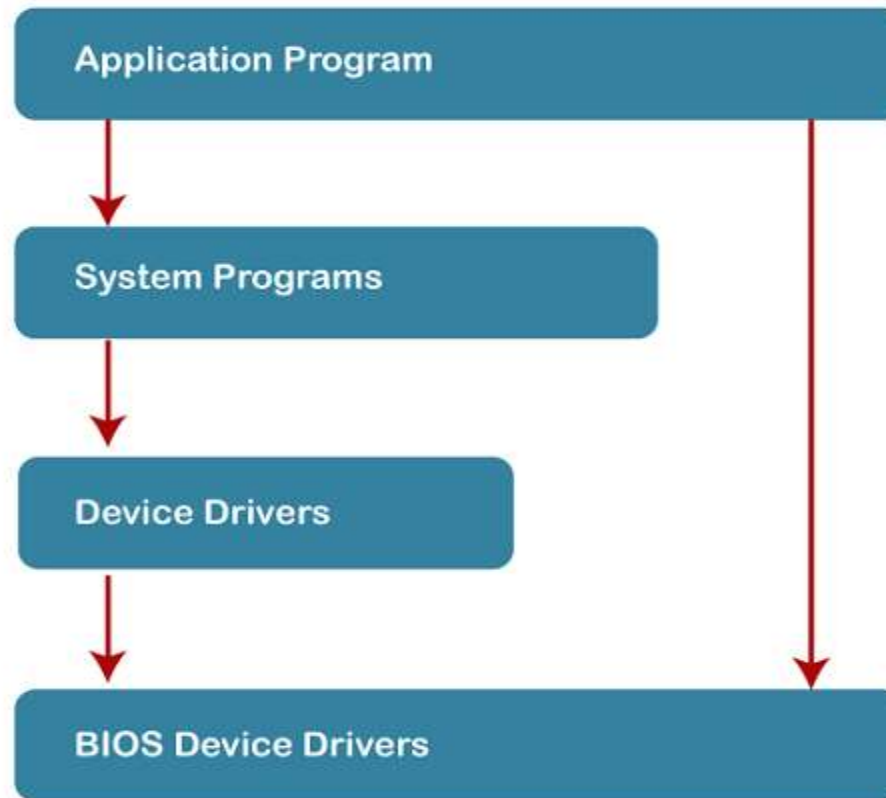
Operating System Structure

- Simple Structure
- Monolithic Structure
- Layered Approach Structure
- Micro-Kernel Structure
- Exo-Kernel Structure
- Virtual Machines

SIMPLE STRUCTURE

- It is the most straightforward operating system structure, but it lacks definition and is only appropriate for usage with tiny and restricted systems
- Four layers: ROM BIOS device drivers, MS-DOS device drivers, application programs, and system programs.

SIMPLE STRUCTURE



Simple Structure Diagram

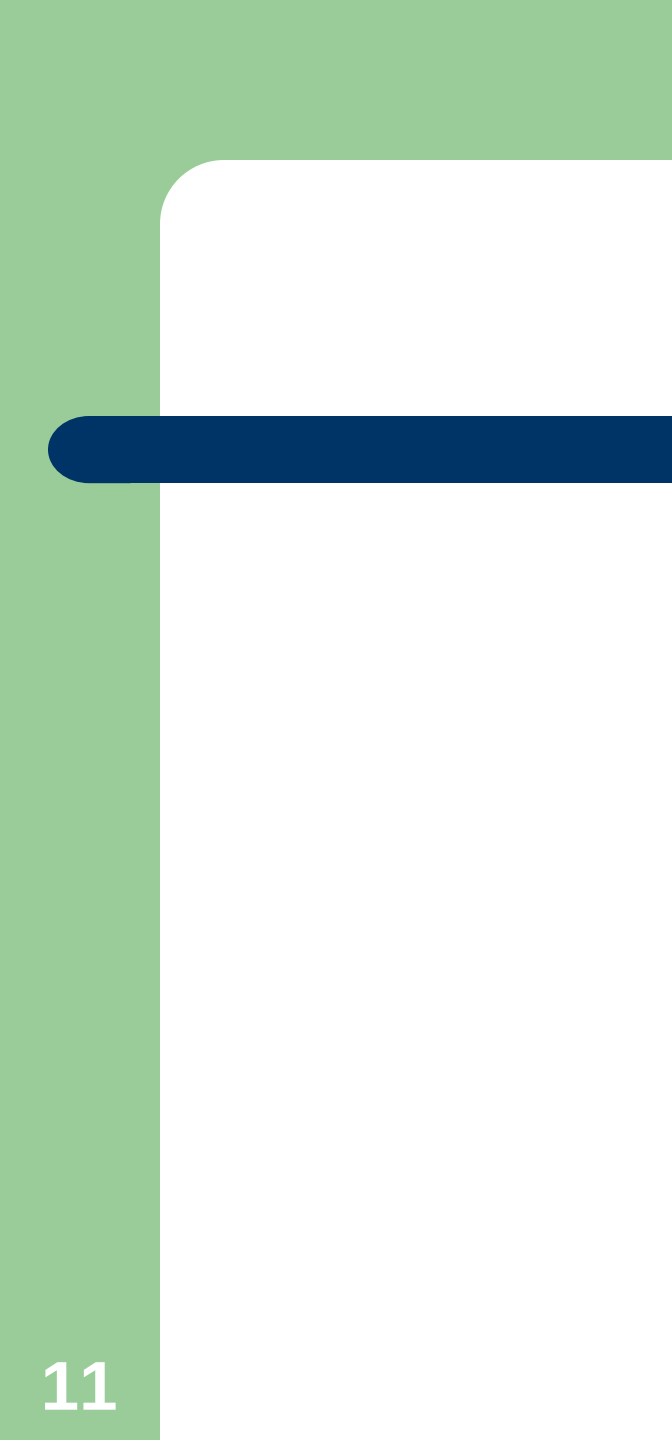
SIMPLE STRUCTURE

Advantages:

- Because there are only a few interfaces and levels, it is simple to develop.
- Because there are fewer layers between the hardware and the applications, it offers superior performance.

Disadvantages:

- The entire operating system breaks if just one user program malfunctions.
- Since the layers are interconnected, and in communication with one another, there is no abstraction or data hiding.
- The operating system's operations are accessible to layers, which can result in data tampering and system failure.



Thank You